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Introduction lecture

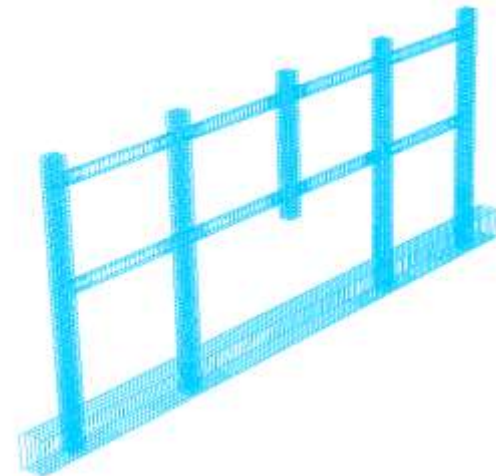
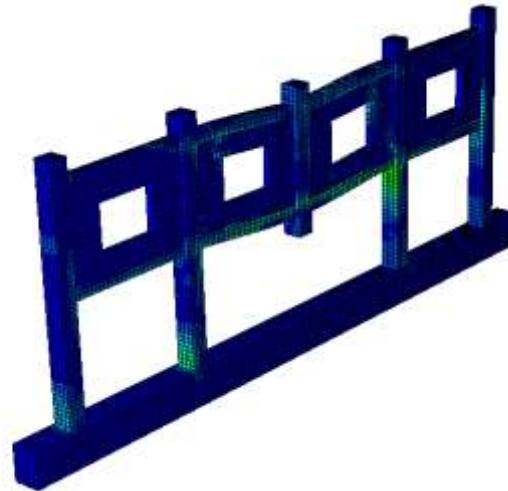
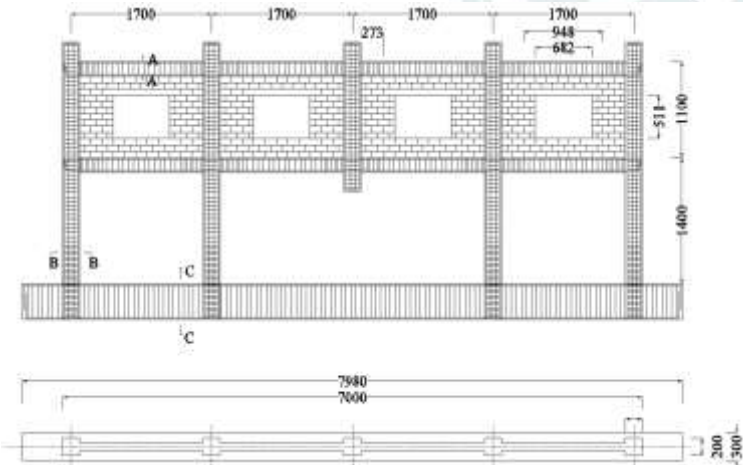
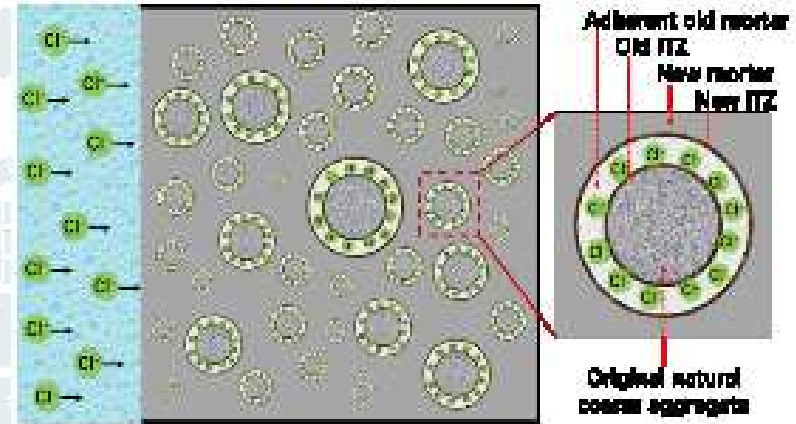
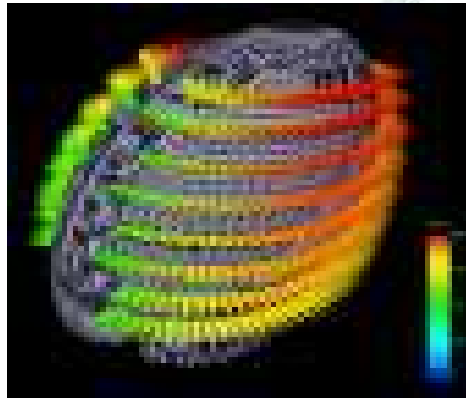




What is FEM

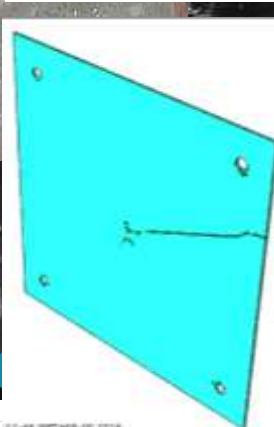
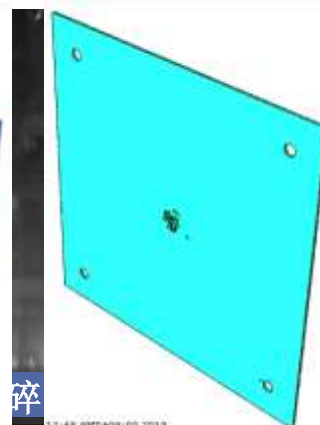
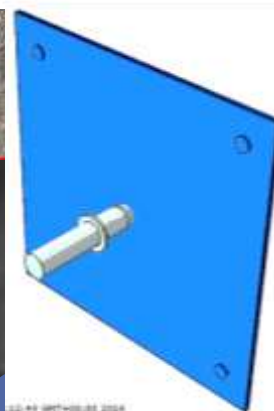
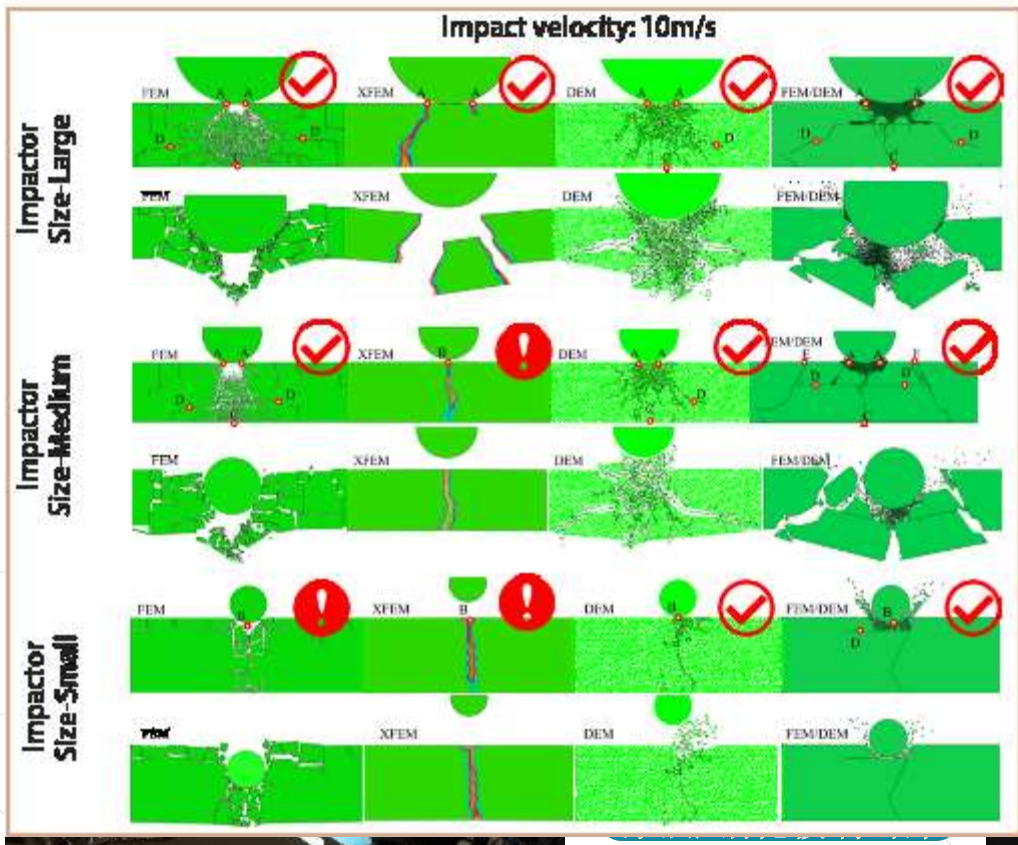


FEM, also called as finite element analysis (FEA), is a method for numerical solution of field problems.





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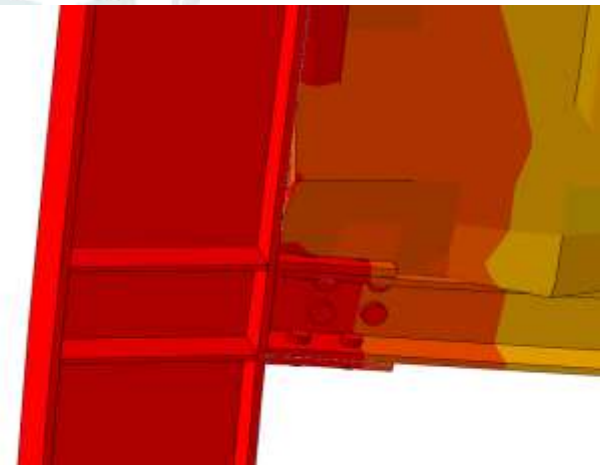
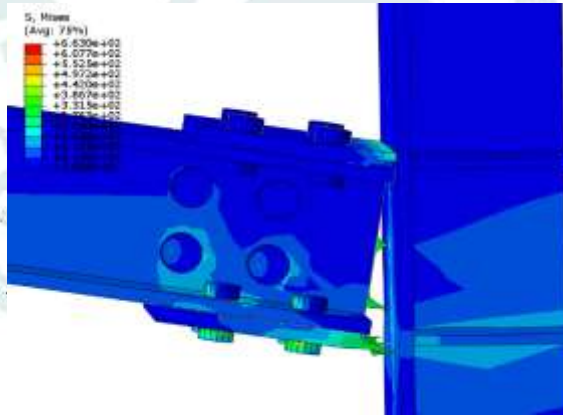
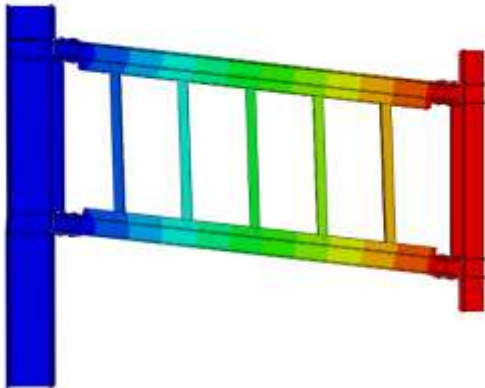




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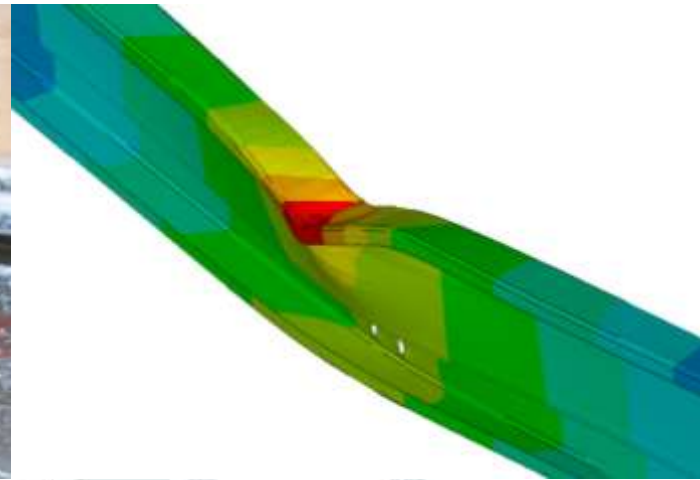


FE modelling



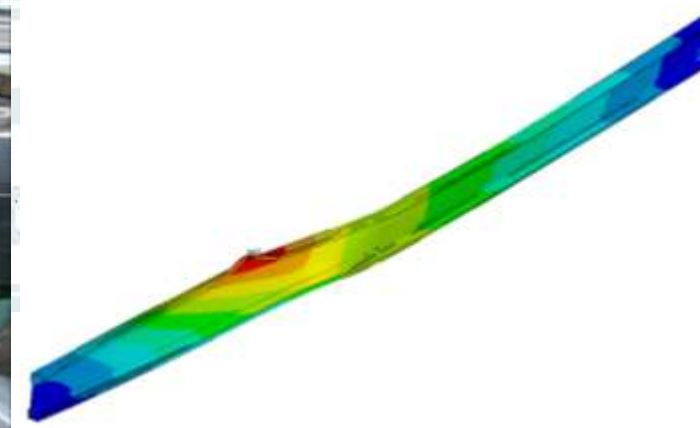
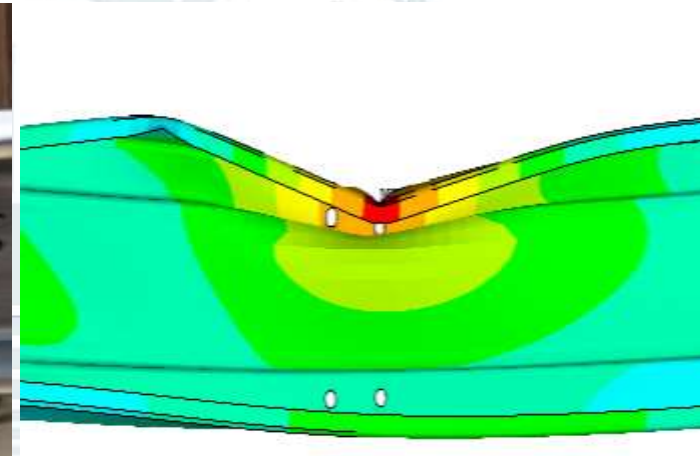


SHANGHAI
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SCIENCE AND
TECHNOLOGY



1987

2006





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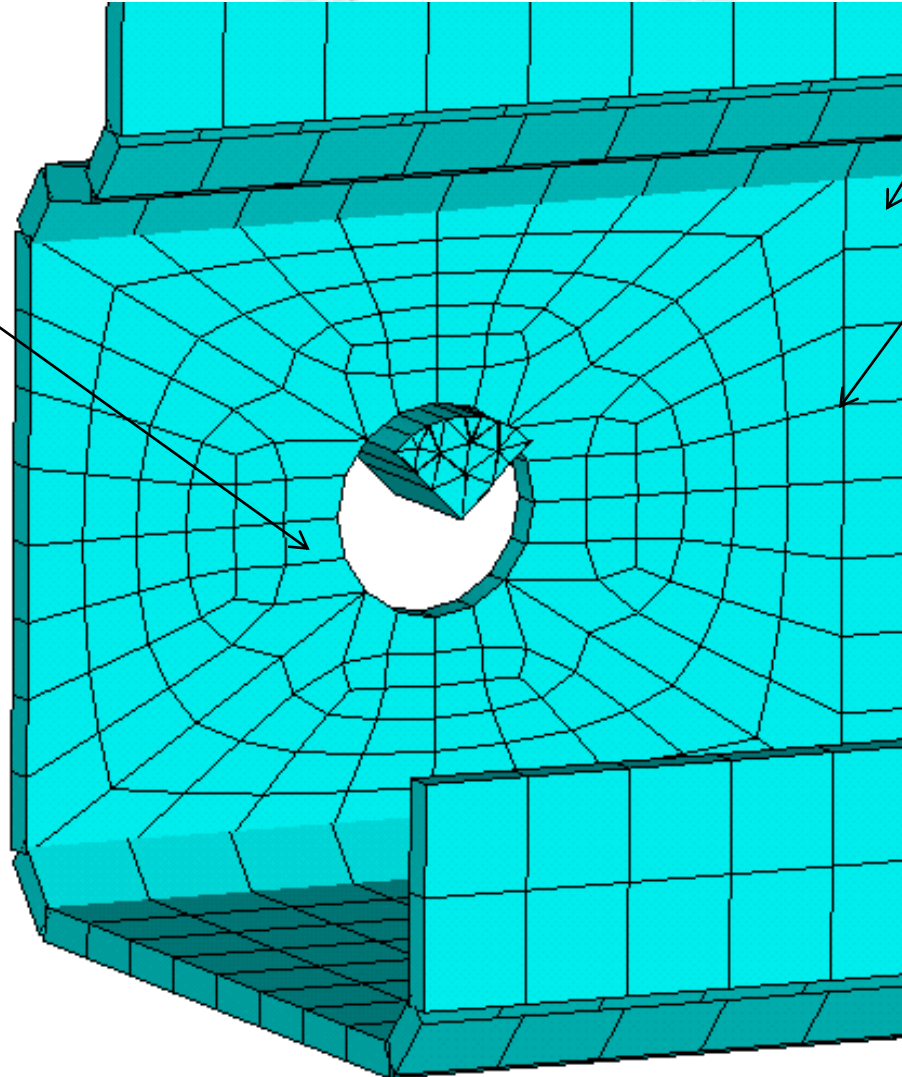
Discretisation



Element

Node

Mesh





Alternative numerical methods

- Other numerical solution methods:
 - **Finite differences**
 - » Approximates the derivatives in the differential equation using difference equations.
 - » Useful for solving heat transfer and fluid mechanics problems.
 - » Works well for two-dimensional regions with boundaries parallel to the coordinate axes.
 - » Cumbersome when regions have curved boundaries.
 - **Weighted residual methods**
 - » Collocation
 - » Subdomain
 - » Least squares*
 - » Galerkin's method*
 - **Variational methods***



Advantages of FEA

- Versatile
- No geometric restriction
- Boundary conditions and loading are not restricted
- Material properties are not restricted
- Components that have different behaviours
- FE models can closely resembles the actual body
- Approximation is easily improved by refining the mesh or increasing the order of elements
- Can deal with multi-physics problems.



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Terminologies

- Interpolation
- Elements
- Nodes
- D.O.F
- ○ ○



History of FEA

- 1943 – Courant (Variational method)
- 1956 (1954) – Turner, Clough, Martin and Topp (Stiffness)
- 1955 – Argyris
- Clough, Argyris, Zienkiewicz the three founders of FEA; *K. Feng made great contributions but was less known.*
- 1960– Clough (proposed Finite Element, plane problems)
- 1970s – Applications on mainframe computers
- 1980s – Microcomputers, pre- and post-processors
- 1990s – Analysis of large structural systems



Available commercial FEM software

- ANSYS
- SDRC/I-DEAS (complete CAD/CAM/CAE package)
- Nastran
- ABAQUS
- COSMOS
- ALGOR
- PATRAN
- HyperMesh
- LS-Dyna
- Comsol



Solving a problem by FEA

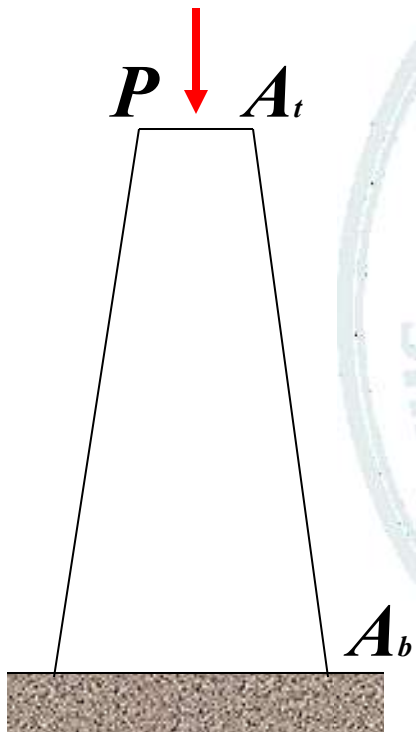
- Defining problems
- Model idealization
- Preliminary analysis
- Finite element analysis
 - Preprocessing
 - Numerical analysis
 - Postprocessing
- Check and interpret results
- Reiteration



E.g.

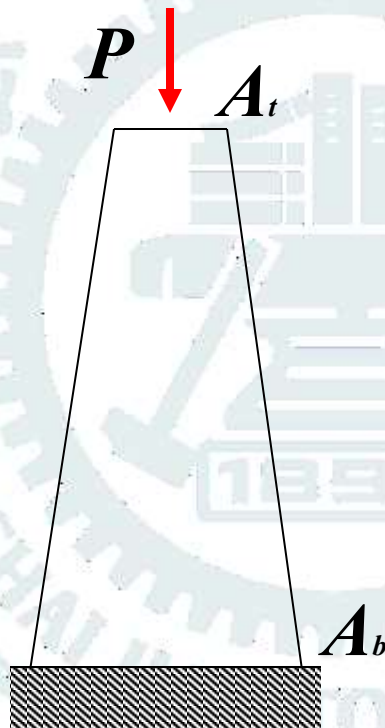


Structure



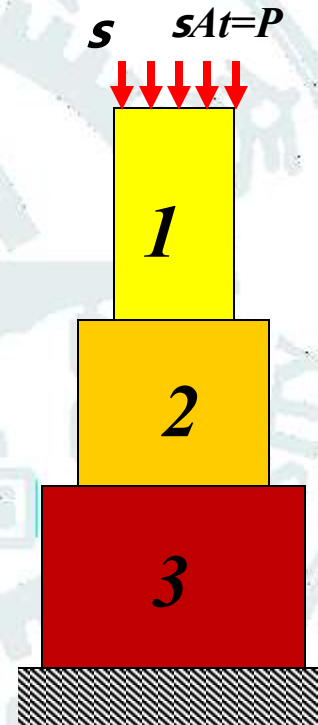
Ground

Model

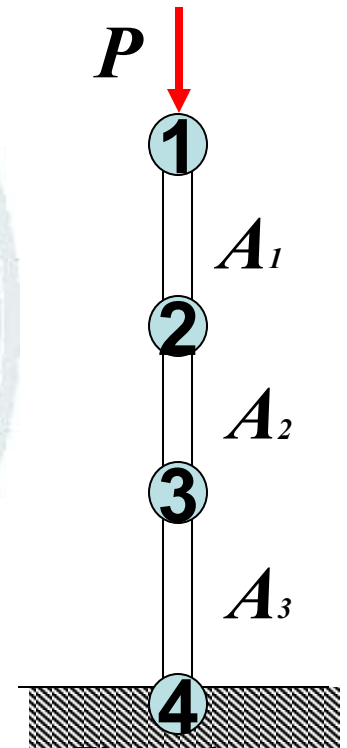


*Rigid
support*

Discretized models



*Physical
representation*



*Finite element
representation*



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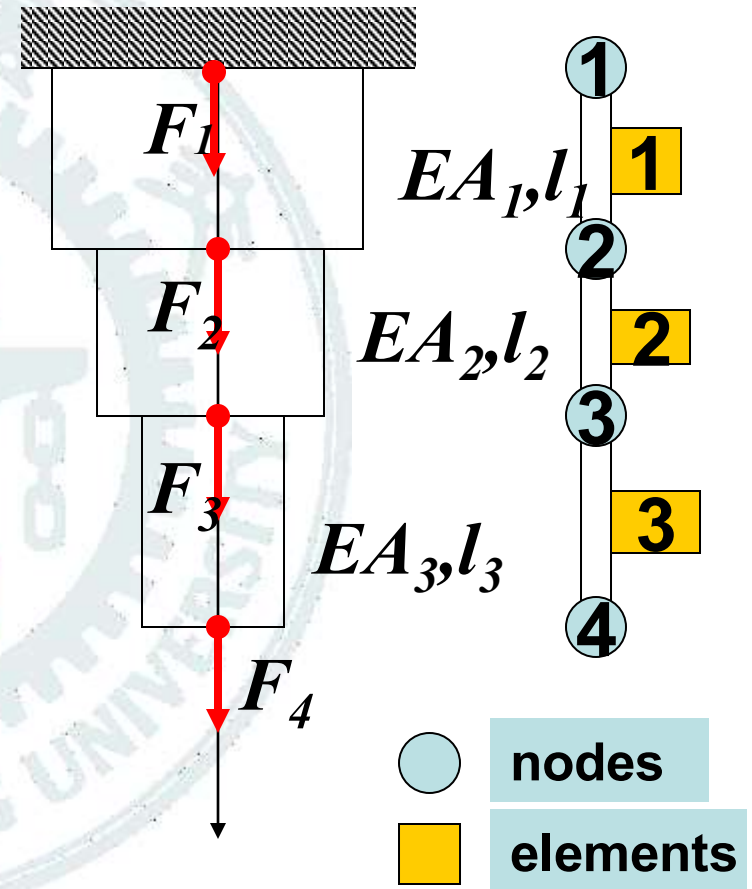
Types of finite elements

- 1-D (line) element
- 2-D (plane) element
- 3-D (solid) element



How does the FEA work - example

Consider the bar of different cross-sections subjected to three point loads as shown in the figure. Determine the displacements at the loading points.





Element stiffness matrices

Equilibrium equation
for element 1

$$\begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} P_{11} \\ P_{12} \end{Bmatrix}$$

$$k_i = \frac{EA_i}{l_i}$$

Equilibrium equation
for element 2

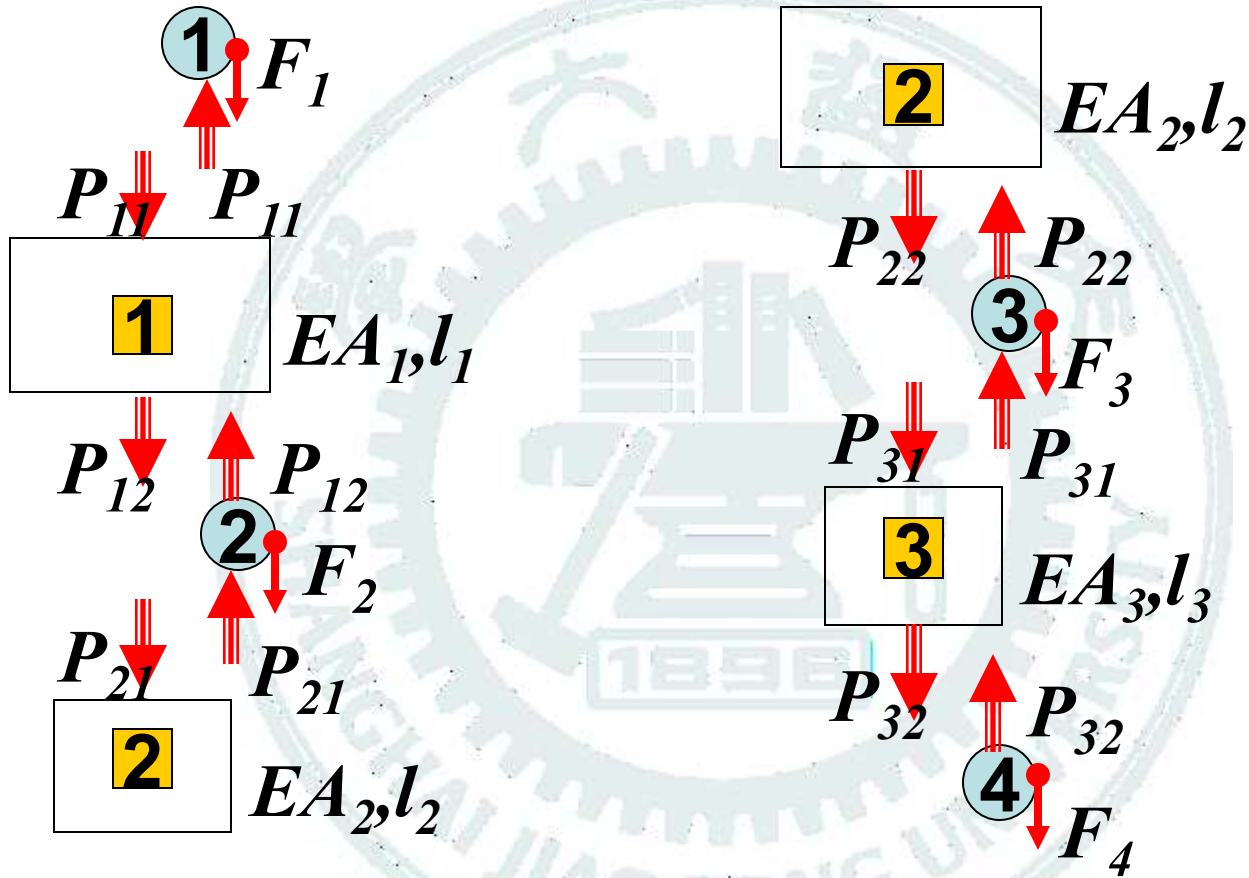
$$\begin{bmatrix} k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} P_{21} \\ P_{22} \end{Bmatrix}$$

Equilibrium equation
for element 3

$$\begin{bmatrix} k_3 & -k_3 \\ -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} P_{31} \\ P_{32} \end{Bmatrix}$$



Equilibrium of nodes



$$\begin{aligned} P_{11} &= F_1 \\ P_{12} + P_{21} &= F_2 \end{aligned}$$

$$\begin{aligned} P_{22} + P_{31} &= F_3 \\ P_{32} &= F_4 \end{aligned}$$



Assembly of stiffness matrices

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} P_{11} \\ P_{12} \\ 0 \\ 0 \end{Bmatrix} \quad \text{For element 1}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ P_{21} \\ P_{22} \\ 0 \end{Bmatrix} \quad \text{For element 2}$$

Element equilibrium equations in the global coordinate system



Assembly of stiffness matrices

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ P_{31} \\ P_{32} \end{Bmatrix} \quad \text{For element 3}$$

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} P_{11} \\ P_{12} + P_{21} \\ P_{22} + P_{31} \\ P_{32} \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$



Applying boundary conditions

$$u_1 = 0, \quad F_1 = R$$

Stiffness matrix

Global equilibrium equations

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} 0 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} R \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$

Solve for nodal displacements from the global equilibrium equations

Solve for reaction forces from the global equilibrium equations

Calculate the strain for each element from obtained displacements

Calculate the stress for each element from obtained displacements



Properties of stiffness matrices

- Nonnegative K_{ii}
- Symmetry
- Sparsity
- Singularity (no support or being mechanism)



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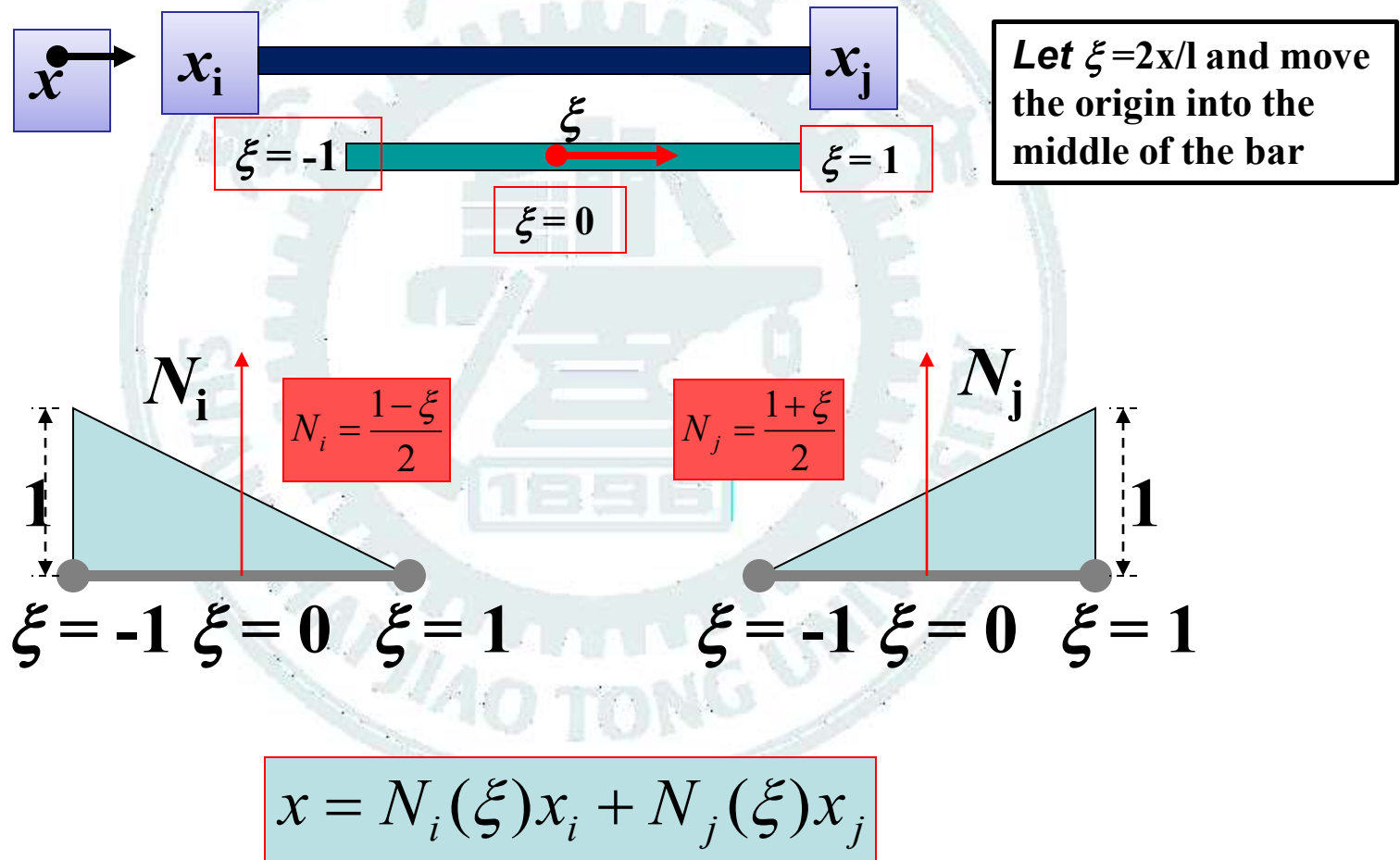


Checking results

- Deformed shape of the structure
- Balance of the external forces
- Order of magnitudes of results
- ...



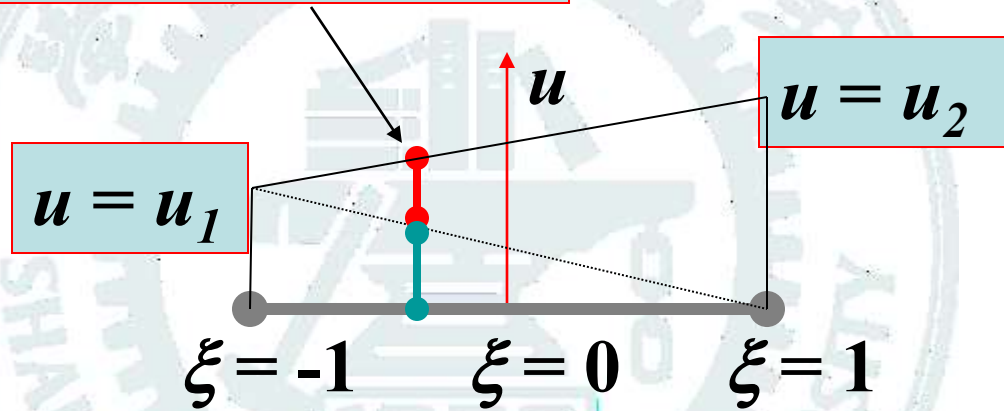
A genetic approach to formulate stiffness matrix





Element displacement field

$$u(x) = N_i(\xi)u_1 + N_j(\xi)u_2$$



$$u = \begin{bmatrix} N_i & N_j \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$



Element strain field

$$\varepsilon(x) = \frac{du}{dx} = \left[\frac{dN_i}{dx} \quad \frac{dN_j}{dx} \right] \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$\frac{dx}{d\xi} = \frac{dN_i}{d\xi} x_i + \frac{dN_j}{d\xi} x_j = -\frac{x_i}{2} + \frac{x_j}{2} = \frac{x_j - x_i}{2} = \frac{l}{2}$$

$$\frac{dN_i}{dx} = \frac{dN_i}{d\xi} \frac{d\xi}{dx} = \left(-\frac{1}{2} \right) \left(\frac{2}{l} \right)$$

$$\frac{dN_j}{dx} = \frac{dN_j}{d\xi} \frac{d\xi}{dx} = \left(\frac{1}{2} \right) \left(\frac{2}{l} \right)$$

$$\varepsilon = \begin{bmatrix} -\frac{1}{l} & \frac{1}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$



Element stress field

$$\sigma(x) = E\varepsilon(x) = E \begin{bmatrix} \frac{dN_i}{dx} & \frac{dN_j}{dx} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$\sigma = \begin{bmatrix} -\frac{E}{l} & \frac{E}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

Note: element strain and element stress are constant



Element internal force

$$N(x) = \sigma(x)A = EA \begin{bmatrix} \frac{dN_i}{dx} & \frac{dN_j}{dx} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$N = \begin{bmatrix} -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$



Element strain energy

$$\begin{aligned} U &= \frac{1}{2} \int_0^V \sigma \varepsilon dV = \frac{Al}{2} \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{bmatrix} -\frac{E}{l} \\ \frac{E}{l} \end{bmatrix} \begin{bmatrix} -\frac{1}{l} & \frac{1}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} \end{aligned}$$

$$[K] = \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix}$$

Element stiffness matrix



Work in the element



$$\begin{aligned} W &= -\int_l q u dx - (P_i u_i + P_j u_j) \\ &= -q \int_l \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} N_i \\ N_j \end{Bmatrix} dx - (P_i u_i + P_j u_j) \\ &= -\begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} \frac{ql}{2} \\ \frac{ql}{2} \end{Bmatrix} - \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} P_i \\ P_j \end{Bmatrix} \end{aligned}$$



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Principle of minimum potential energy for a single element

$$\Pi = U + W = \frac{1}{2} \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} - \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} \frac{ql}{2} \\ q\frac{l}{2} \end{Bmatrix} - \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} P_i \\ P_j \end{Bmatrix}$$

$$\frac{\partial \Pi}{\partial \mathbf{u}^T} = \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} - \begin{Bmatrix} \frac{ql}{2} \\ q\frac{l}{2} \end{Bmatrix} - \begin{Bmatrix} P_i \\ P_j \end{Bmatrix} = 0$$

$$\begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} = \begin{Bmatrix} \frac{ql}{2} + P_i \\ q\frac{l}{2} + P_j \end{Bmatrix}$$

Element equilibrium equation



$$\mathbf{k}\mathbf{u} = \mathbf{f}$$

where

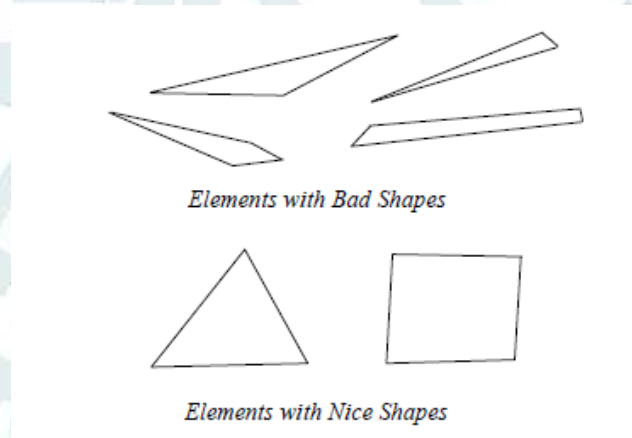
$$\mathbf{k} = \int_V (\mathbf{B}^T E \mathbf{B}) dV$$

is the *element stiffness matrix*.



Choosing elements

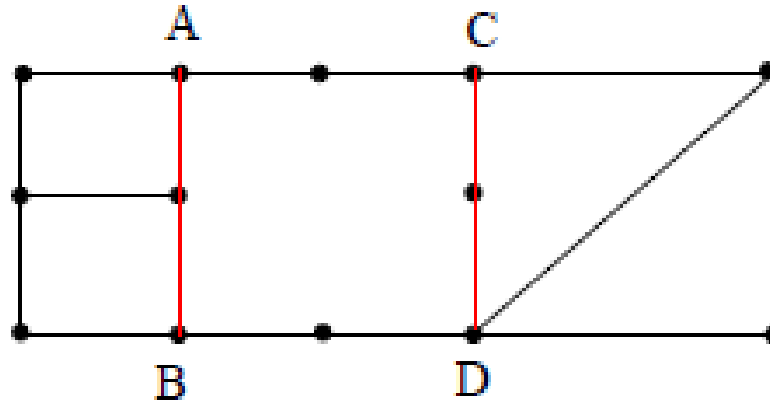
- Choose the right type of element for a given problem
- Avoid elements with large aspect ratios and corner angle



- Connect elements correctly



Is the following meshing correct?





Modeling and solution techniques

- Use symmetry
 - Reflective symmetry
 - Rotational (cyclic symmetry)
 - Axisymmetry
- Substructures
- Convergence of FE solutions
 - Reduce the element size
 - Increase the order of element



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Global dynamic equations

$$[\mathbf{M}]\{\ddot{\mathbf{D}}\} + [\mathbf{C}]\{\dot{\mathbf{D}}\} + [\mathbf{K}]\{\mathbf{D}\} = \{\mathbf{R}^{\text{ext}}\}$$



Explicit integration

$$\frac{1}{\Delta t^2} [\mathbf{M}] \{\mathbf{D}\}_{n+1} = \{\mathbf{R}^{\text{ext}}\}_n - \left[\mathbf{K} - \frac{2}{\Delta t^2} \mathbf{M} + \frac{1}{\Delta t} \mathbf{C} \right] \{\mathbf{D}\}_n - \left[\frac{2}{\Delta t^2} \mathbf{M} - \frac{1}{\Delta t} \mathbf{C} \right] \{\mathbf{D}\}_{n-1}$$



Implicit integration

$$\begin{aligned} [\mathbf{K}^{\text{eff}}] \{\mathbf{D}\}_{n+1} = & \{\mathbf{R}^{\text{ext}}\}_{n+1} + [\mathbf{M}] \left[\frac{1}{\beta \Delta t^2} \{\mathbf{D}\}_n + \frac{1}{\beta \Delta t} \{\dot{\mathbf{D}}\}_n + \left(\frac{1}{2\beta} - 1 \right) \{\ddot{\mathbf{D}}\}_n \right] \\ & + [\mathbf{C}] \left[\frac{\gamma}{\beta \Delta t} \{\mathbf{D}\}_n + \left(\frac{\gamma}{\beta} - 1 \right) \{\dot{\mathbf{D}}\}_n + \Delta t \left(\frac{\gamma}{2\beta} - 1 \right) \{\ddot{\mathbf{D}}\}_n \right] \end{aligned}$$

where

$$[\mathbf{K}^{\text{eff}}] = \frac{1}{\beta \Delta t^2} [\mathbf{M}] + \frac{\gamma}{\beta \Delta t} [\mathbf{C}] + [\mathbf{K}]$$

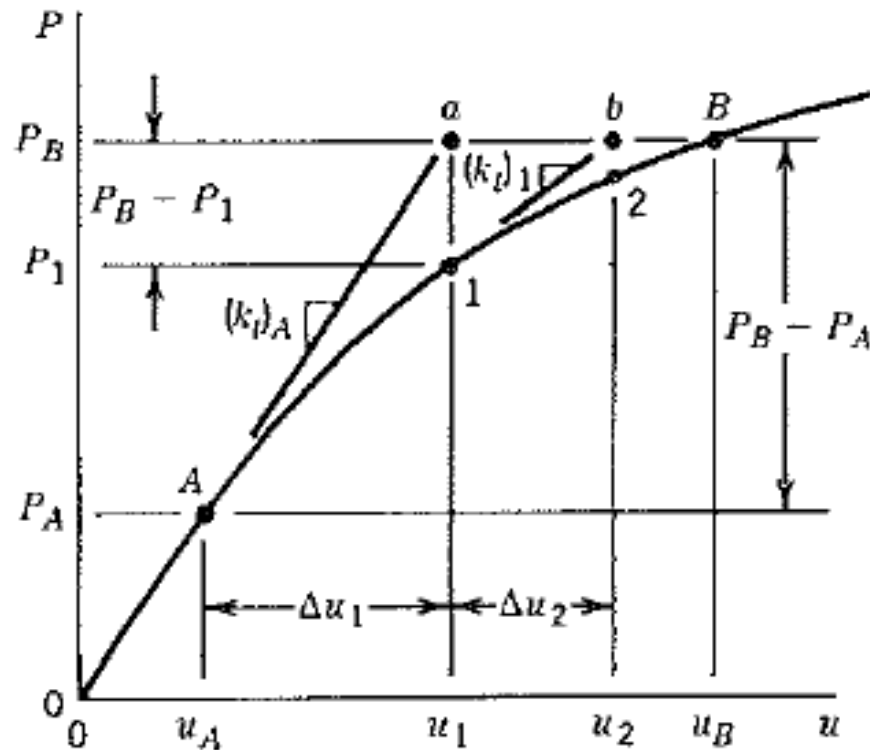


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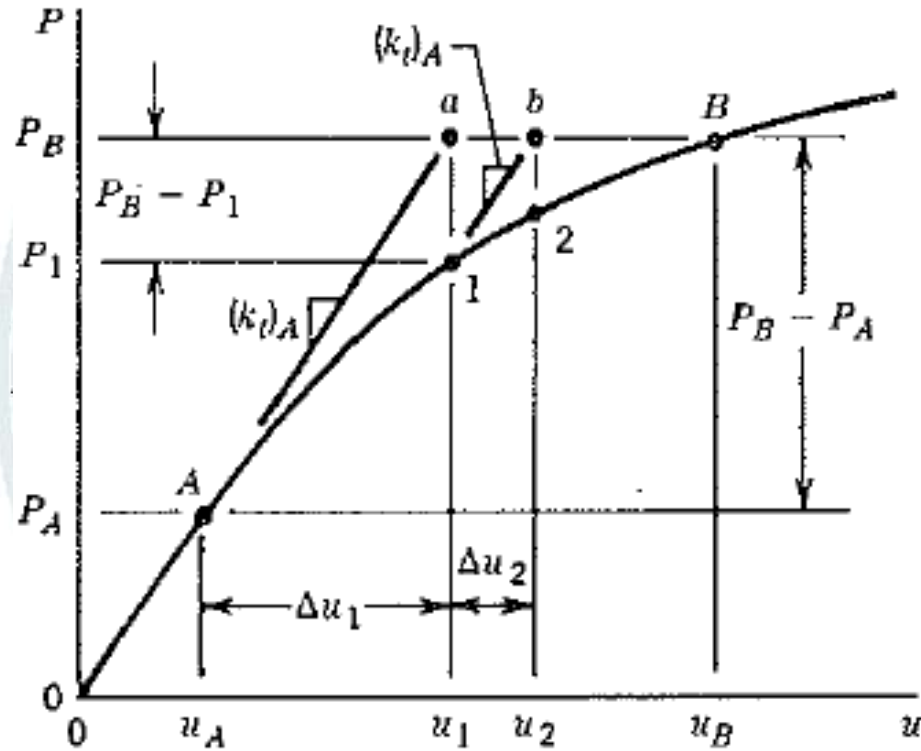
Nonlinear problems

- Material nonlinearity
- Contact nonlinearity
- Geometric nonlinearity



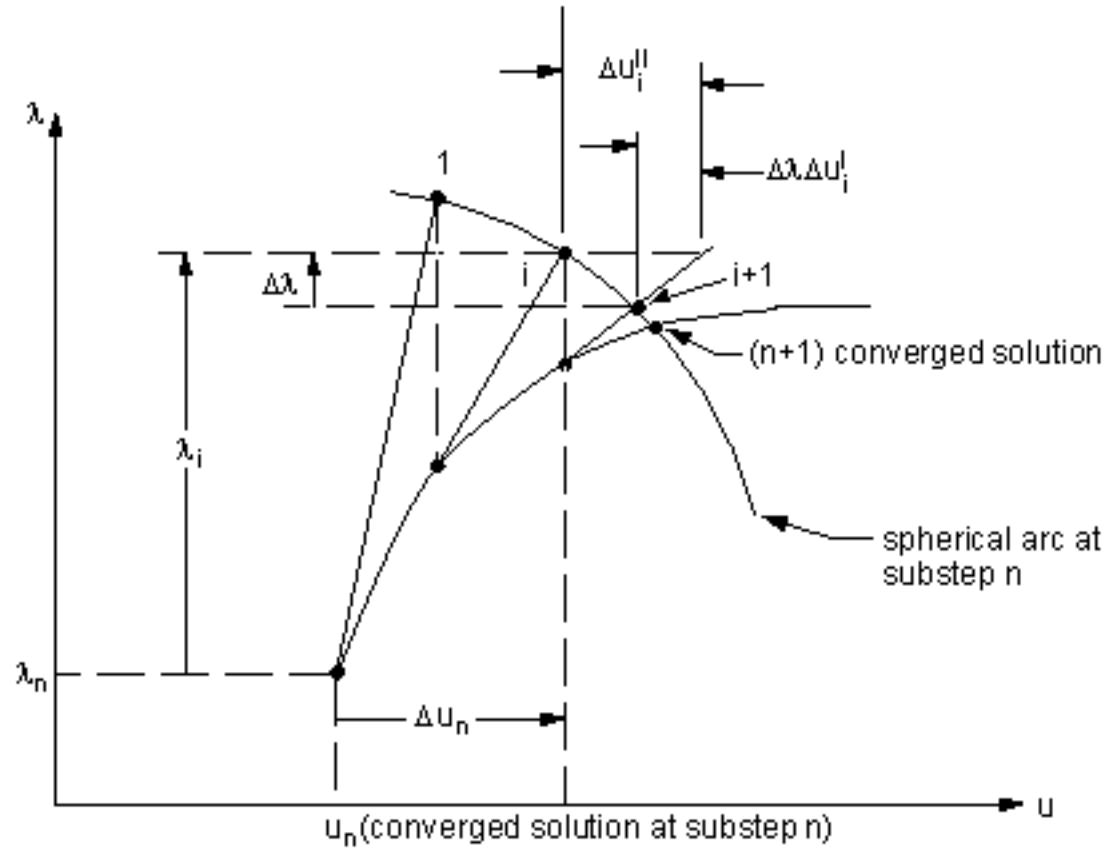


Modified Newton-Raphson iteration





Arc-length method





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- Rao, S.S. The finite element method in engineering (5th ed.) Elsevier Butterworth-Heinemann. 2011.
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See more details about the course in the
Course Syllabus file in Canvas.



Thanks !

